

An Empirical Study on Multi-Paradigm AI Text Detection: Generalization Gaps, Explainability, and LLM-as-a-Judge Failures

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Abstract

As generative AI approaches human parity in text generation, the necessity for robust detection mechanisms has surged. This paper presents a comprehensive comparative analysis of four distinct text classification paradigms: Traditional Machine Learning (Logistic Regression), Deep Sequence Models (LSTMs), Fine-Tuned Transformers (BERT), and Generalized LLM-as-a-Judge (Qwen/Phi). We evaluate these architectures on accuracy, inference latency, and memory footprint. Our findings reveal that while BERT achieved the highest test accuracy (94.4%), zero-shot generalized LLMs performed worse than simple baseline models (38%). Furthermore, our explainability audit using LIME exposed critical vulnerabilities across all paradigms, including the "Length Paradox," "Sigmoid Saturation" in LSTMs, and the "Generalization Gap" in Transformers. To bridge the gap between research and production, we deployed these models via a full-stack MLOps pipeline featuring a FastAPI backend, an SQLite logging system, and a Streamlit dashboard. The complete source code, datasets, and Streamlit application for this study are available at <https://github.com/23f2001127/read-between-the-lines>

1. Introduction

The proliferation of sophisticated Large Language Models (LLMs) has initiated an "arms race" in digital forensics. Modern AI generation relies heavily on Reinforcement Learning from Human Feedback (RLHF) to minimize the stylistic markers that early detectors used to flag synthetic text.

This study moves beyond simple accuracy metrics by auditing how different AI paradigms "think." We investigate whether generalized, instruction-tuned LLMs can detect AI text out-of-the-box, how sequence models handle padding, and why highly accurate Transformers fail in real-world scenarios. Finally, we demonstrate how these models can be successfully integrated into a production-ready application.

2. Methodology & Architectural Paradigms

The study evaluates text classification across four computational paradigms, trained and tested on a stratified dataset of human and AI-generated texts.

2.1 Baseline Machine Learning (TF-IDF + Logistic Regression)

To establish a computational floor, we implemented a Logistic Regression model utilizing TF-IDF (Term Frequency-Inverse Document Frequency) vectorization. This "Bag of Words" approach ignores grammatical sequence and relies entirely on statistical vocabulary distribution.

2.2 Deep Sequence Models (LSTM)

To capture the sequential context missed by TF-IDF, we trained a Long Short-Term Memory (LSTM) network.

- **Architectural Fix (Post-Padding):** During the engineering phase, we identified a critical preprocessing vulnerability. Pre-padding sequences (inserting zeros before the text) caused the

LSTM's hidden state to "wash out" before reaching the actual input. Transitioning to **post-padding** ensured the model processed meaningful tokens while its memory state was fresh, stabilizing convergence and restoring accuracy from 71% back to ~90%.

2.3 State-of-the-Art Transformers (BERT)

We fine-tuned bert-base-uncased (110M parameters). BERT's bidirectional self-attention mechanism processes the entire sequence simultaneously, allowing it to capture deep semantic relationships. To optimize GPU VRAM utilization during evaluation, a custom dynamic mini-batching pipeline was engineered.

2.4 Generalized LLM-as-a-Judge

We evaluated efficient instruction-tuned models (e.g., Qwen 1.5B / Phi) without fine-tuning to test their out-of-the-box detection capabilities. We utilized two prompting strategies: Zero-Shot classification and Persona Chain-of-Thought (CoT).

2.5 Production MLOps Integration

To evaluate these models in a real-world context, they were wrapped in a **FastAPI** backend. A **Streamlit** frontend was developed to allow simultaneous multi-paradigm inference, while an **SQLite** database was integrated to log all user queries and model predictions for continuous monitoring and drift detection.

3. Quantitative Results & Evaluation

Models were evaluated on a holdout test set. We measured traditional classification metrics alongside Inference Latency and Model Size to assess deployment viability.

	Accuracy	F1-Score	Precision	Recall	Latency (ms)	Size (MB)
Logistic Regression	0.825594	0.825833	0.825018	0.826649	0.568929	0.153411
LSTM	0.901291	0.903287	0.885707	0.921580	1.118307	10.272761
BERT	0.944516	0.944702	0.941904	0.947517	8.049855	440.039149

3.1 The Failure of Generalized LLMs

A major finding of this study is the poor performance of zero-shot LLMs. Despite their massive parameter counts, models like Qwen performed worse than a simple coin flip (35% to 50% accuracy) and were significantly outperformed by the 0.15 MB Logistic Regression model. This indicates that generalized instruction-tuning does not inherently grant a model the ability to detect synthetic statistical signatures without dedicated fine-tuning.

4. Qualitative Insights & Explainability (LIME)

Relying solely on test-set accuracy creates a false sense of security. We utilized LIME (Local Interpretable Model-Agnostic Explanations) to audit the decision-making processes of our models, revealing several critical vulnerabilities.

4.1 The Length Paradox

Our application testing revealed that AI detection becomes statistically unreliable on texts under 50 words. NLP models require "textural" patterns to detect synthetic signatures. Short, generic greetings

(e.g., "Hi, how are you?") lack sufficient statistical texture, often resulting in erratic or highly skewed predictions.

4.2 The LSTM "God Complex"

LIME analysis revealed that the LSTM frequently outputted extreme confidence intervals (95%+) on both correct and incorrect predictions. This "God Complex" is attributed to **Sigmoid Saturation**. The model over-indexed on strict grammatical perfection, pushing probabilities to extremes at the slightest hint of synthetic structure. Sequence model probabilities must therefore be treated as relative rankings rather than absolute certainties.

4.3 The Transformer Generalization Gap

Despite achieving 94.4% accuracy on the test set, BERT exhibited a notable **Generalization Gap**. When exposed to novel inference data from cutting-edge LLMs (e.g., modern ChatGPT prompts), BERT's performance degraded. The model overfit to the static distribution of its training data and struggled against the evolved "human jitter" introduced by modern RLHF. This proves that detectors are heavily bound by the temporal relevance of their datasets.

5. Conclusion & Recommendations

This empirical study demonstrates that there is no "silver bullet" for AI text detection. While BERT offers state-of-the-art accuracy on known distributions, it is computationally heavy and vulnerable to dataset shift. Conversely, zero-shot LLMs are entirely unsuited for this task despite their advanced reasoning capabilities.

For real-world production systems, we recommend a **Hybrid Cascade Architecture**:

1. Route incoming text through a lightweight Logistic Regression model for rapid, low-cost baseline screening.
2. Utilize deep learning (LSTM/BERT) only when the text length is sufficient and the baseline confidence requires secondary verification.

Future work will focus on implementing continuous active learning pipelines within our SQLite database to periodically retrain the BERT model, mitigating the Generalization Gap as generative AI continues to evolve.